

JAVA PROGRAM REPORT

Hierarchical Inheritance in Java

Programming Language: Java

Topic: Hierarchical Inheritance

Program File: Hyrarchical.java

Objective

To demonstrate hierarchical inheritance in Java, where two different derived classes, Pizza and Burger, inherit from the same base class, Food.

Java Source Code

```
package core_java;

class Food {
    private String name;

    public Food(String name) {
        this.name = name;
    }

    public String getName() {
        return this.name;
    }

    public void eat() {
        System.out.println("Eating " + this.name + ".");
    }
}

class Pizza extends Food {
    private String topping;

    public Pizza(String name, String topping) {
        super(name);
        this.topping = topping;
    }

    public void bake() {
        System.out.println("Baking " + getName() + " with " + this.topping + ".");
    }
}

class Burger extends Food {
    private boolean hasCheese;

    public Burger(String name, boolean hasCheese) {
        super(name);
        this.hasCheese = hasCheese;
    }

    public void grill() {
        System.out.println("Grilling the " + getName() + " patty.");
    }
}

public class Hyrarchical {
    public static void main(String[] args) {
        Pizza myPizza = new Pizza("Pizza", "Pepperoni");
        myPizza.eat();
        myPizza.bake();

        System.out.println();

        Burger myBurger = new Burger("Burger", true);
        myBurger.eat();
        myBurger.grill();
    }
}
```

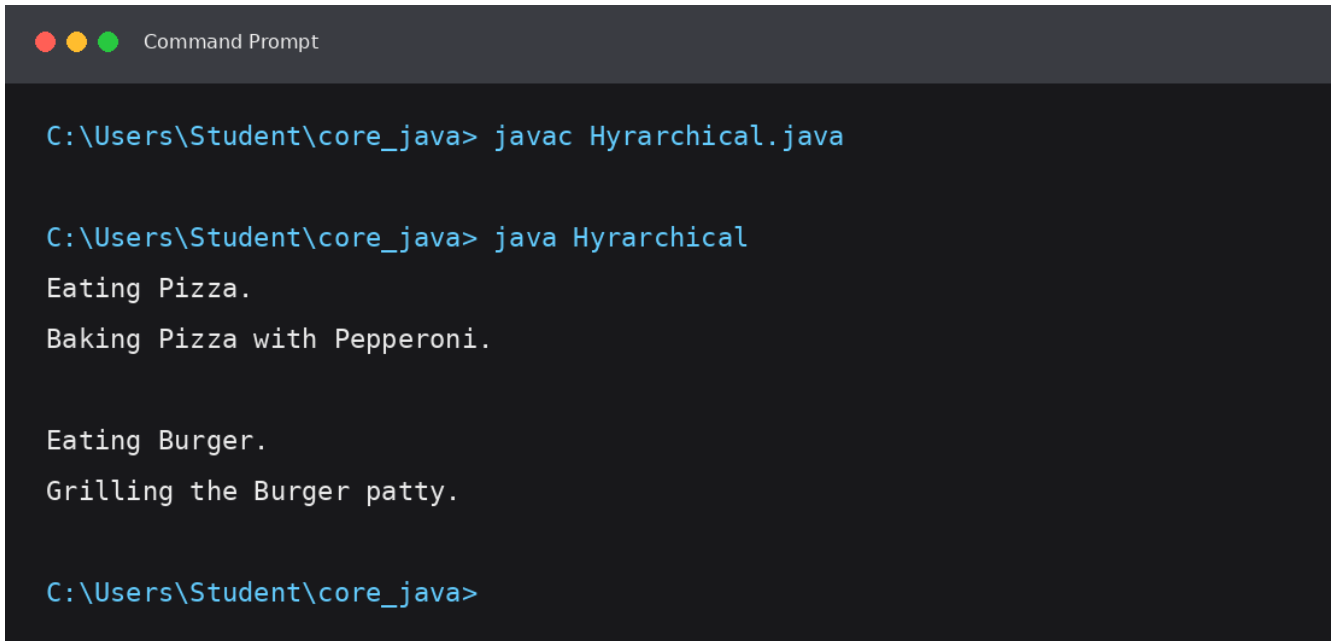
Program Explanation

- 1. Food Class:** This is the parent class containing the name property, getName() method, and eat() method.
- 2. Pizza Class:** Pizza extends Food and inherits its properties and methods. It adds a topping property and bake() method.
- 3. Burger Class:** Burger also extends Food. It adds the hasCheese property and grill() method.
- 4. Hierarchical Inheritance:** Both Pizza and Burger inherit from the same parent class Food. This structure is called hierarchical inheritance.
- 5. Main Method:** Separate Pizza and Burger objects are created and their inherited as well as individual methods are executed.

Execution and Output Screenshots

The compilation screenshot uses the requested file-name format: **javac Hyrarchical.java**.

Test Case 1: Compilation and Execution



```
Command Prompt

C:\Users\Student\core_java> javac Hyrarchical.java

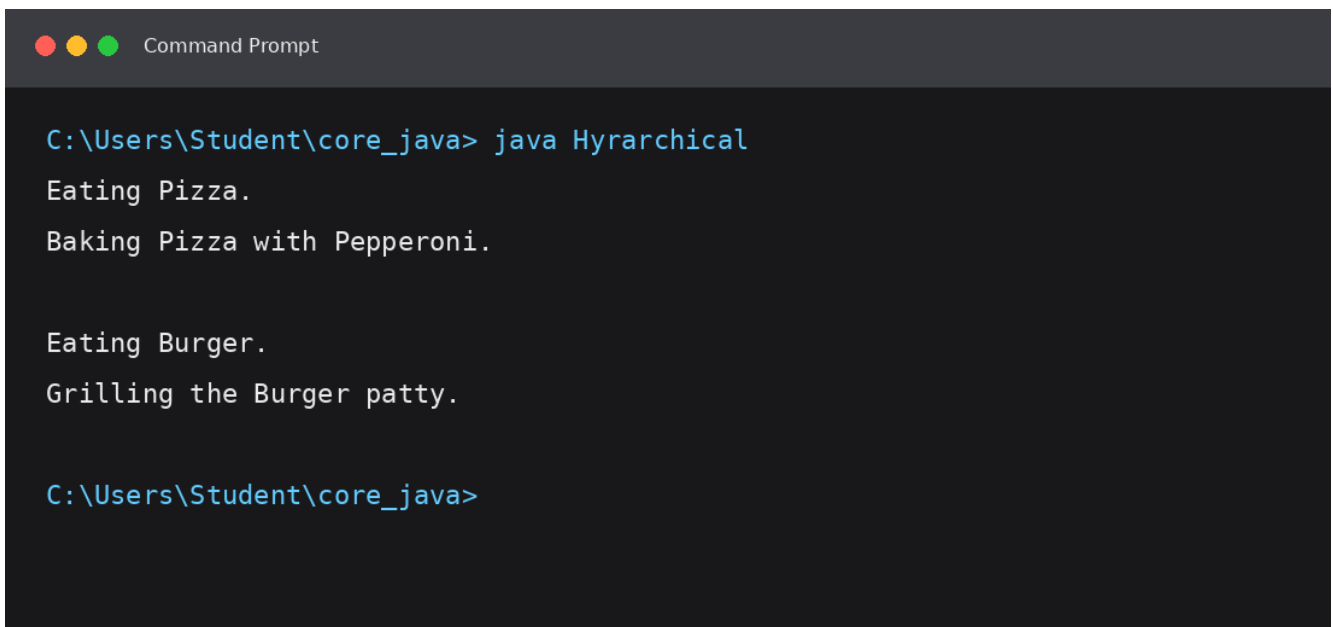
C:\Users\Student\core_java> java Hyrarchical
Eating Pizza.
Baking Pizza with Pepperoni.

Eating Burger.
Grilling the Burger patty.

C:\Users\Student\core_java>
```

Verification: Program compiled successfully using the file name command `javac Hyrarchical.java` and produced the expected output.

Test Case 2: Re-execution Verification



```
Command Prompt

C:\Users\Student\core_java> java Hyrarchical
Eating Pizza.
Baking Pizza with Pepperoni.

Eating Burger.
Grilling the Burger patty.

C:\Users\Student\core_java>
```

Verification: The program was executed again and the output remained consistent, confirming correct hierarchical inheritance behavior.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	Pizza and Burger methods execute correctly	Output matched	PASS
Test Case 2	Same result on re-execution	Output matched	PASS

Conclusion

The program successfully demonstrates hierarchical inheritance. Both Pizza and Burger inherit common functionality from the Food class while maintaining their own specialized methods. The program was compiled and tested successfully.